



ANNO DOMINI MMXXVI

# Locksmithing 101

## Pin-Tumbler Locks



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ANNO DOMINI MMXXVI

# Locksmithing 101 Pin-Tumbler Locks

Presented by Jeremy Reeder

Security Engineer

Master Safecracker

Counterintelligence Advisor

# First Lock in History – 2000 BC

- Ancient Egypt
  - Made of fig wood



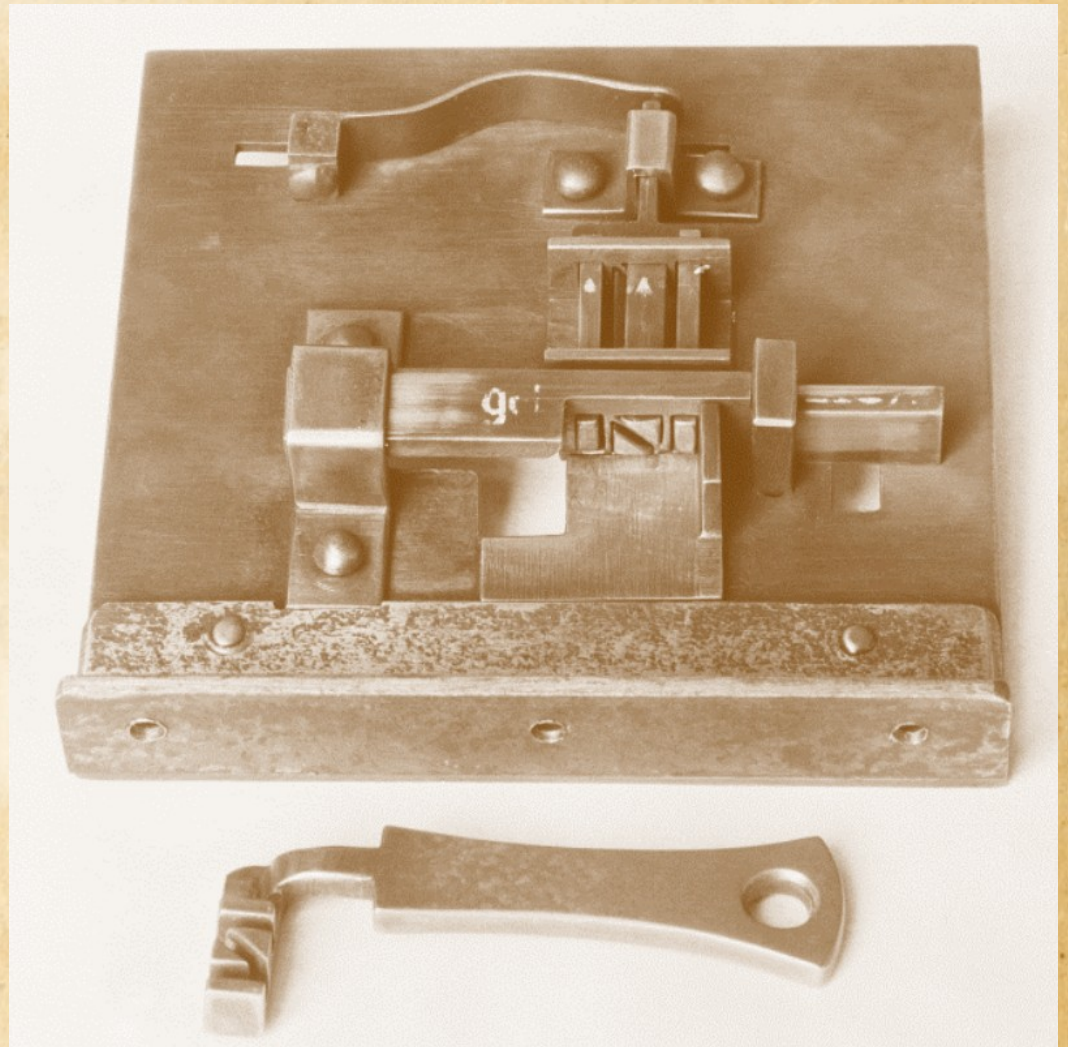
# First Lock in History – 2000 BC

- Ancient Egypt
  - Made of fig wood
  - Used to secure:
    - Homes
    - Shops
    - Tombs (occasionally)



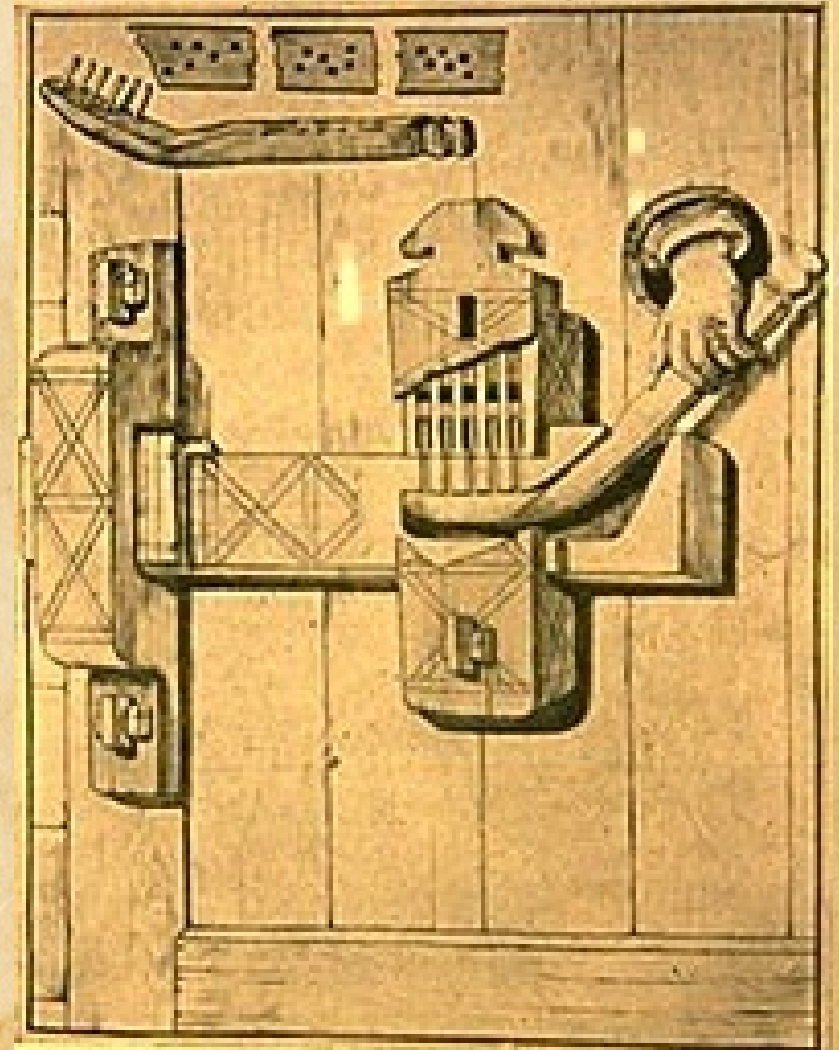
# Door & Gate Locks of Early Roman Empire

- 15% bronze locks
  - Used in palaces
  - Built by redsmiths



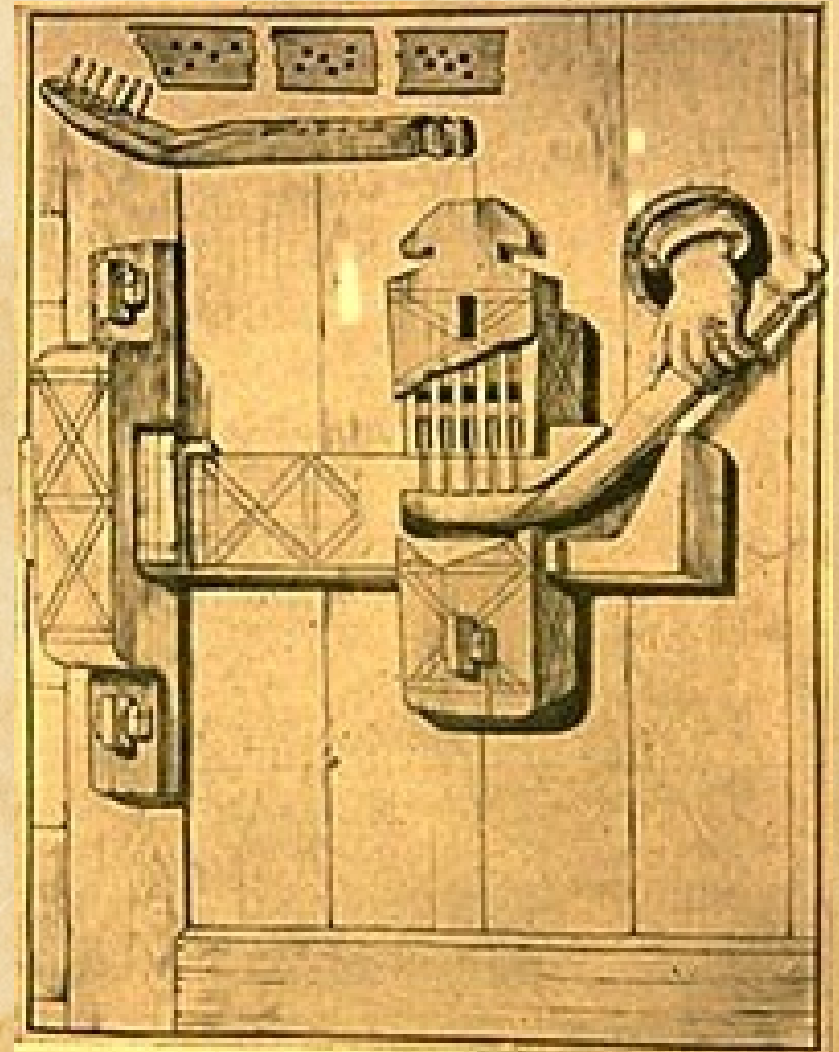
# Door & Gate Locks of Early Roman Empire

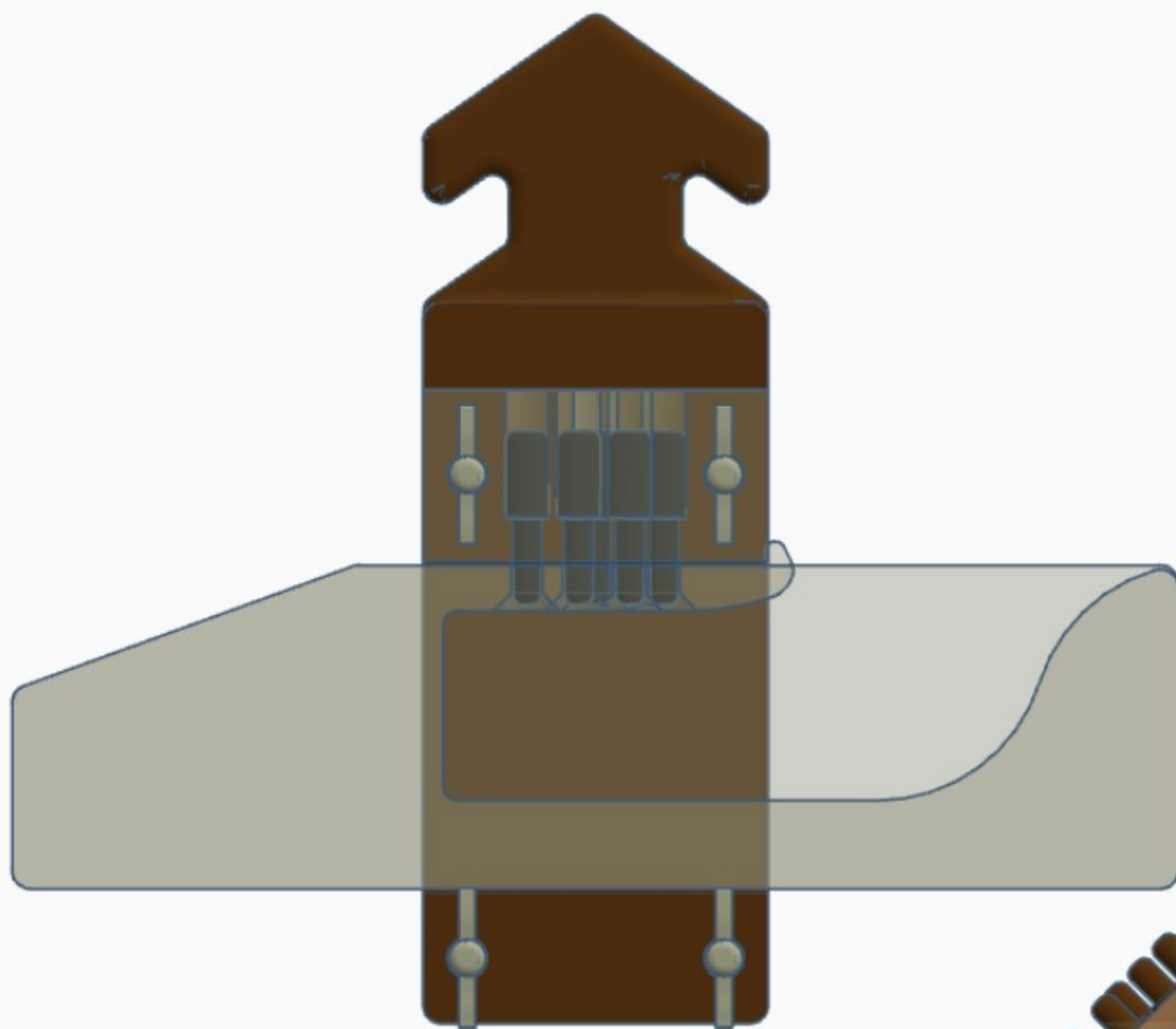
- 15% bronze locks
  - Built by redsmiths
- 85% hardwood locks
  - Tamarisk, oak, olive, walnut



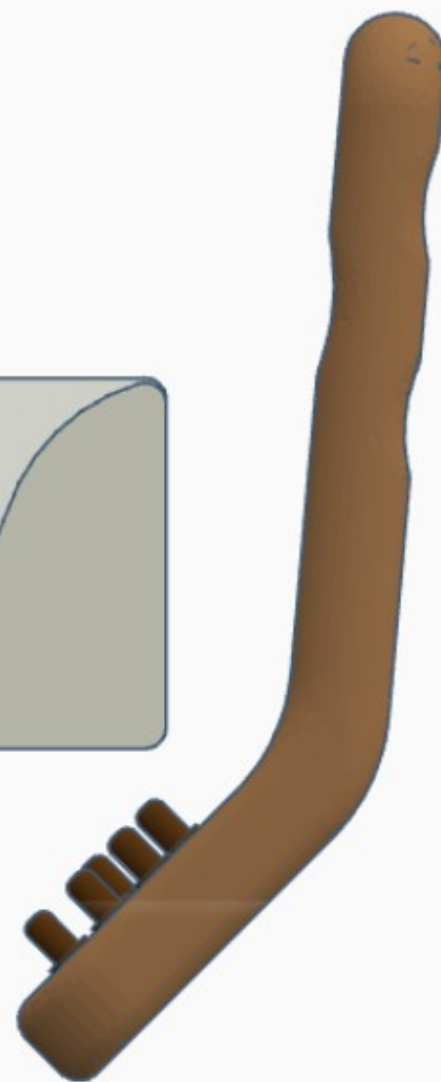
# Door & Gate Locks of Early Roman Empire

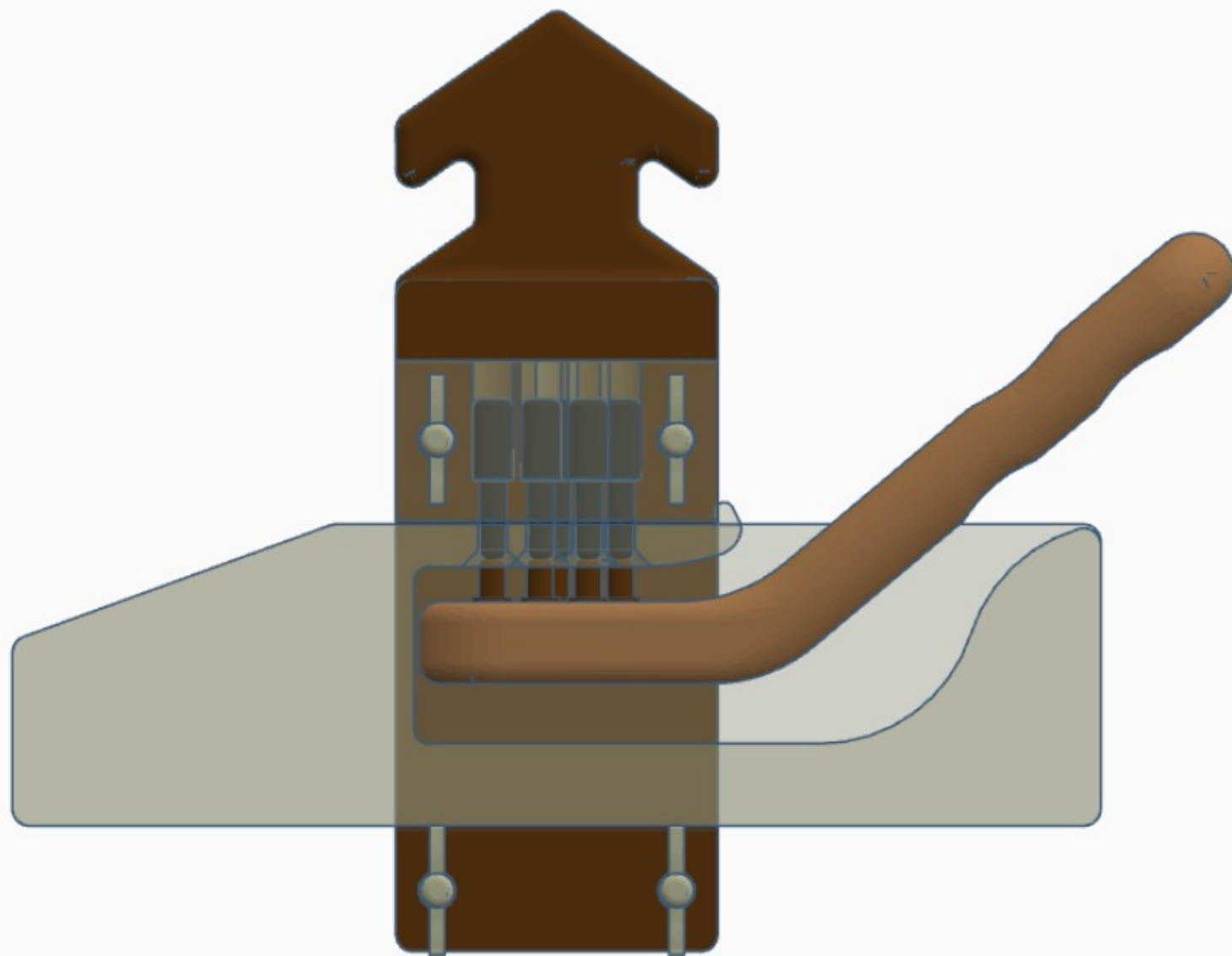
- 15% bronze locks
  - Built by redsmiths
- 85% hardwood locks
  - Tamarisk, oak, olive, walnut
  - Built by carpenters
    - Joseph?
    - Jesus?



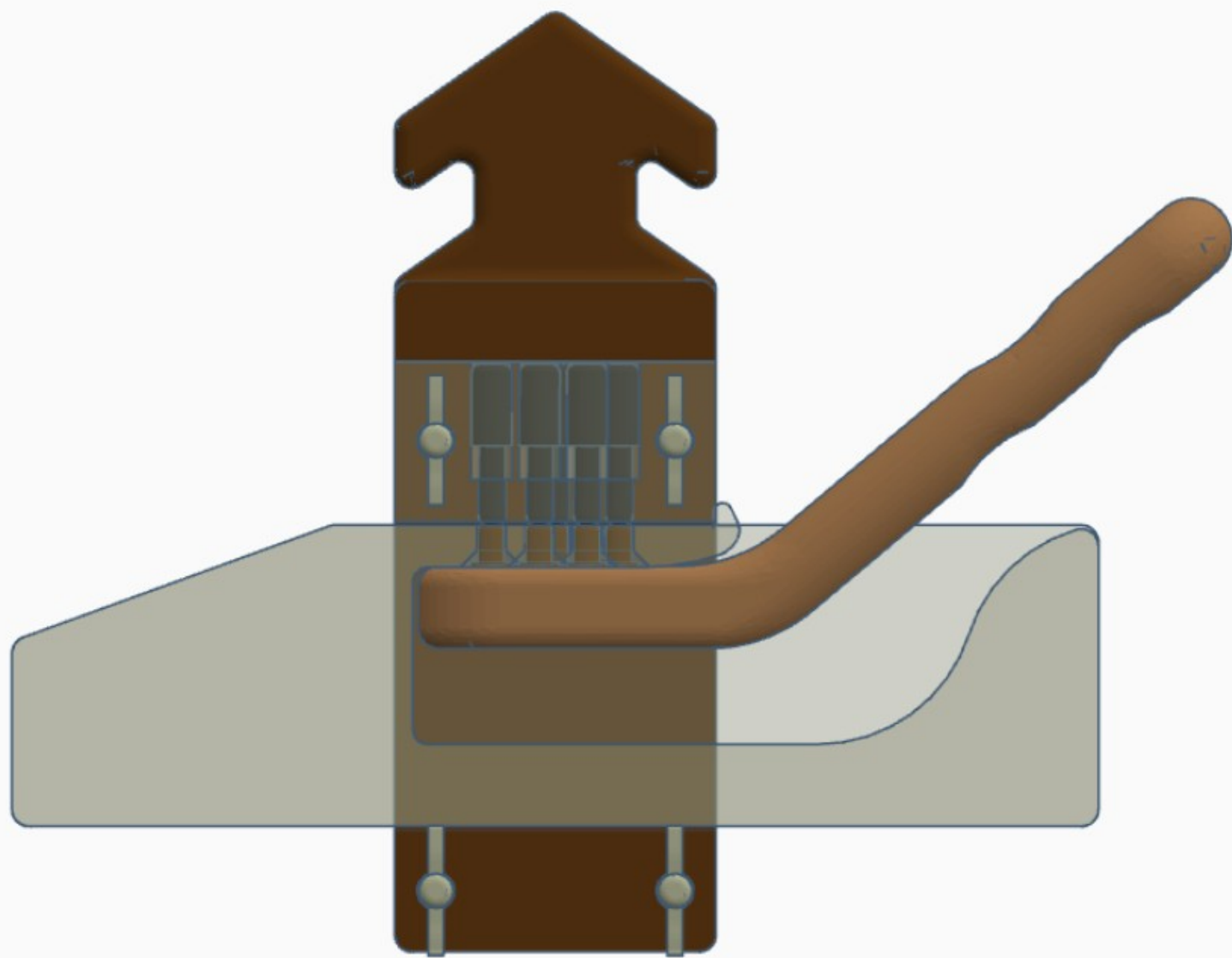


**LOCKED**

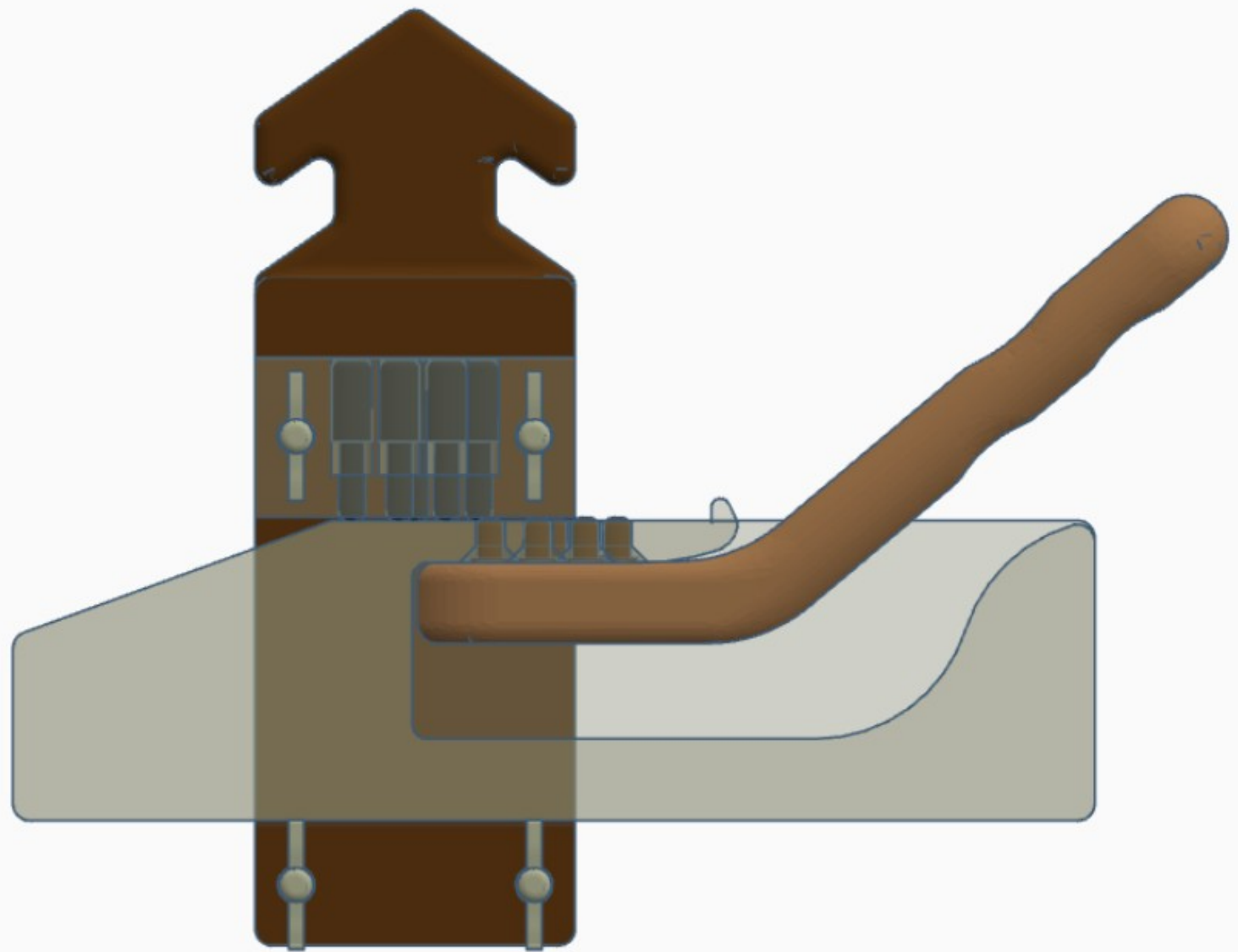




**INSERTING KEY**



**PINS DISENGAGED**



**RETRACTING BOLT**

# Lockpicking Law & Ethics

## Idaho Code § 18-1406 (1972)

POSSESSION OF BURGLARIOUS INSTRUMENTS. Every person having upon him ... a picklock ... or other instrument ... with intent feloniously to break or enter into any building or who shall knowingly make or alter ... any ... instrument above named ... without being requested so to do by some person having the right to open the same, or who shall make ... any instrument ... to be used in committing a misdemeanor or felony, is guilty of a misdemeanor.

## THE CREDO OF THE LOCK PICKER

*YOU MAY ONLY PICK LOCKS THAT YOU OWN, OR  
THOSE TO WHICH YOU'VE BEEN GIVEN EXPLICIT  
PERMISSION TO PICK BY THE RIGHTFUL OWNER*

(Locksport International. *The LSI Guide to Lock Picking*. Page 1)

# What would Indiana do?

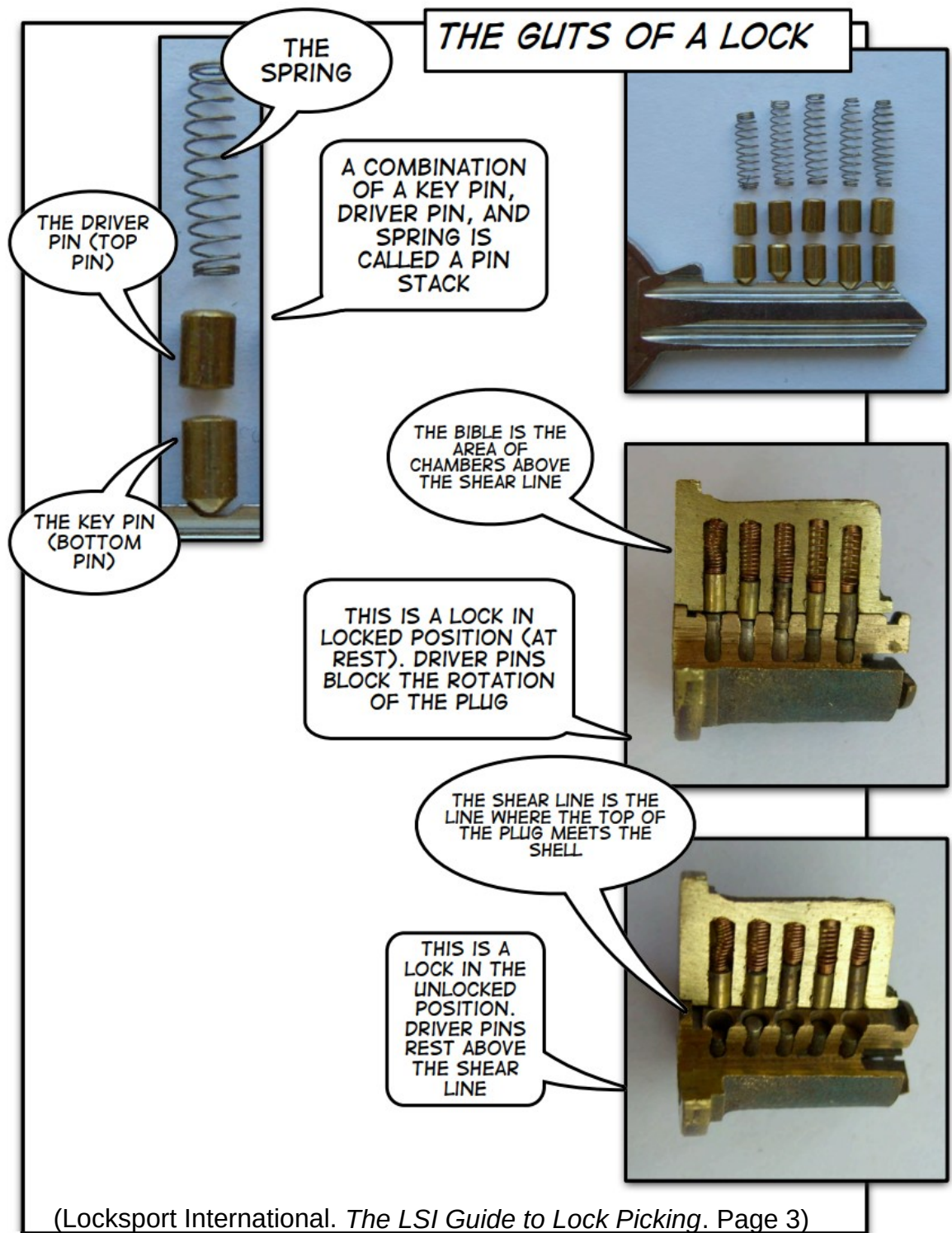


VS



## Modern lock materials:

- soft brass
- hardened brass
- stainless steel
- case-hardened steel
- zamak



# Lock Bypass, Part 1

## Schrödinger's Lock

- What's so dead about a deadbolt?
- What in the world is a deadlatch?
- How do we find out if a lock is alive or dead?

# Lock Bypass, Part 2

## Under the Door



# Lock Bypass, Part 3

## Exploiting an Exit Device



# To Be Continued...

Ideas for Next Week:

- Car Lock Bypass
- Intermediate Lockpicking
- 19<sup>th</sup> Century Bank Vaults